

Chapter 94 — Upper Urinary Tract Trauma — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

1. **Contrast CT (computed tomography) with 10-min delays** = gold standard, kidney and ureter
2. **Image blunt:** gross haematuria, micro + **SBP (systolic BP) <90**, or deceleration
3. **Image penetrating:** any haematuria
4. **Micro, no shock, no deceleration** → observe, no imaging
5. **Operate:** unstable despite resuscitation; **expanding/pulsatile haematoma**
6. Stable grade I–IV (± select V) → **nonoperative**, any mechanism
7. Urinary extravasation → **observe**; 76–90% heal
8. **Revascularise only** solitary kidney or bilateral artery injury
9. **Proximal to iliacs** → ureteroureterostomy over stent; **distal** → reimplant ± hitch
10. **No bowel interposition/autotransplant acutely**

RENAL INJURIES

Epidemiology:

- Kidney = **commonest injured urological organ**; up to 5% of trauma
- **Blunt 80–90%**; commonest blunt cause **motor vehicle collision**
- Penetrating: **firearms** ~2%; kidney in ~10–20% penetrating abdominal wounds
- **Most injuries AAST (American Association for the Surgery of Trauma) I–III**
- Commonest associated organs: **liver, spleen**
- **Ribs 9–12 fracture** → strongest rib association

Mechanism:

- Deceleration → **hilum + UPJ (ureteropelvic junction):** thrombosis, avulsion
- High-velocity bullet >2000 ft/s (610 m/s) → cavitation, delayed necrosis
- **Anterior axillary stab** → hilum; **posterior axillary** → parenchyma

Haematuria:

- Microhaematuria = >5 RBC/hpf; hypotension = **SBP <90** any time
- Degree ≠ severity; ~50% **UPJ injuries no haematuria**
- Use **first urine aliquot** (later = diluted)
- Dipstick sens/spec >97%; **microscopy = gold standard**
- False +ve dipstick: **myoglobin, dehydration, exercise, menses, povidone-iodine**

Grading (AAST 2018):

- II vs III: laceration ≤1 cm vs >1 cm
- III: contained vascular injury within Gerota's
- IV: **collecting system, pelvis/UPJ, segmental vessel**, bleeding beyond Gerota's
- V: **main vessel**, hilar avulsion, shattered, devascularised + bleeding
- **Advance one grade** for multiple injuries, **up to III**

Children:

- Image **shock + micro** (>5 RBC/hpf); liberal imaging
- BP maintained until ~50% **blood volume lost**

CT:

- Arterial ~80 s → bleeding; delayed → urine leak; optimal delay **9 min**
- Red flags: **medial haematoma** (pedicle), **medial extravasation** (UPJ), **no enhancement/cortical rim** (artery)
- Grade IV high risk: ≥2 of rim >4 cm, medial laceration, vascular extravasation
- **MiGUTS (Multi-Institutional Genito-Urinary Trauma Study)** nomogram: rim distance biggest; vascular extravasation 29 pts
- **CT misses renal vein injury**

- **FAST (focused sonography):** poor renal specificity; **can't tell blood from urine**
- **Urinoma 0–20 HU; haematoma >30 HU; abscess >20 HU + rim**

One-shot IVP (IV pyelogram):

- **2 mL/kg IV push, single film at 10 min**
- Purpose: **functioning contralateral kidney**
- Alternative: clamp ipsilateral ureter + **IV methylene blue** → blue urine

Nonoperative:

- Serial Hb **every 4–6 h**; **early mobilisation safe**
- Blunt conservative success >90%
- Fails if: instability, **2–3 units transfused**, new guarding
- Rarely fails <24 h; most failures → **stent, nephrostomy or AE**
- Reimage (AUA) ~48 h grade IV–V deep laceration or symptoms; **not grade I–III**
- Extravasation reimage **48–72 h**
- **Early discharge <48 h** safe: select blunt grade I–III
- Stent impossible (urethral distraction, transection) → **PCN (percutaneous nephrostomy) ± antegrade**
- Delayed bleed: usually **within 2 weeks** (up to 21 days) → **AE**

Operate if (absolute):

- **Unstable**, no/transient response
- **Expanding/pulsatile haematoma**
- **Pedicle avulsion; UPJ avulsion**

Do not operate:

- Stable well-staged I–III at laparotomy → **leave Gerota's closed**
- **Stable zone II haematoma** → observe
- Extravasation alone grade IV → **90% resolve**; >25% devitalised → lower threshold

Angioembolisation (AE):

- **Superselective**; repeat AE highly successful
- AE fails: **instability, grade V, extravasation, gross haematuria**
- Pseudoaneurysm/AVF (arteriovenous fistula) → **AE**

Exploration:

- At laparotomy, Gerota's opened → >½ **nephrectomy**
- **Transabdominal; vessel control before opening Gerota's**
- Incision over aorta **above IMA** (inferior mesenteric artery) → Treitz → left renal vein
- Haematoma → incise **medial to IMV**
- Zones: **I central, II lateral (kidney), III pelvis**
- Damage control: pack, **return 24–48 h**
- Poor survival: **GCS <8, base excess <-13.9**

Vascular:

- Warm ischaemia <120 min
- Main artery repair salvages ~½
- Bilateral: ~½ **avoid dialysis**
- Hypertension after repair ≥43% → BP check **1 year**
- **Left renal vein ligatable** (collaterals); **right not**

Complications:

- **Goldblatt** = arterial stenosis/occlusion
- **Page kidney** = compression → **ACEi/ARB**; refractory → decortication
- Hypertension ↑ after artery occlusion (**32–50%**)
- Mortality high-grade ~**5–8%**; nephrectomy–mortality link **conflicting**
- USANZ (Urological Society of Australia and New Zealand): **US 4–6 wk**, microhaematuria check **6 mo**
- Return to sport **2 wk–3 mo**; rebleed risk **2–6 wk**

URETERAL INJURIES

Epidemiology:

- **1–2.5%** genitourinary trauma; **iatrogenic commonest overall**
- External: **gunshot commonest**; **>90% associated injuries**
- External → **proximal**; iatrogenic → **distal**
- **25–45% no haematuria**
- Mortality **6–9%**

AAST ureter:

- II **<50%**; III **≥50%**; IV transection **<2 cm** devasc; V avulsion **>2 cm**

Iatrogenic:

- **Hysterectomy** = leading cause; **prolapse surgery** highest gynae range
- Lap > open gynae; **lap = open colorectal**
- URS (ureteroscopy) avulsion **<1%**; stricture **0.3–3%**
- Risk order: **EHL > Nd:YAG > Ho:YAG**
- Unstented ureter → sheath **≤14 Fr**
- Perforation → **stop, stent**
- **Endometriosis** ↑ risk; ~**10% women** distal supply via uterine artery
- Devascularisation presents **1 wk–2 mo**: stenosis, urinoma, fistula
- Hysterectomy injury: mean diagnosis **day 10**
- Vascular: **85% not recognised**; stenosis **1–2%**
- Ureteroarterial fistula → haematuria; **chronic stent**; **covered endovascular stent**

Dyes:

- **Methylene blue**: MAOI → avoid **SSRI/SNRI/MAOI, pregnancy, G6PD**; false low SpO₂

- **Indigo carmine**: t_½ **4–5 min**, blue in 5–10 min; hypotension
- **Phenazopyridine** most cost-effective
- ICG (indocyanine green) **5 mL of 2.5 mg/mL** per ureter
- Bilateral stents → anuria **1–5%**

Imaging:

- **CT urogram, 10-min delay** = best (stable)
- **Retrograde** = most sensitive; test for missed injury
- One-shot IVP non-diagnostic **33–100%**
- Missed injury: **fever, leucocytosis, peritonism**; ~11% missed at laparotomy

Repair principles:

- **Spatulated, tension-free, stented, watertight, drained**
- **Non-tunnelled** reimplant in trauma
- Unstable → **ligate (long silk) + delayed PCN**; **no open nephrostomy**
- Damage control: nothing/re-look 24 h, exteriorised stent, cutaneous ureterostomy, ligate + PCN
- Missed contusion → **stent 6 weeks** (~85%)
- Endoscopic injury → **stent and/or PCN**
- **Ureterovaginal fistula** → **stent first**

By level:

- Upper ⅓ → **UU (ureteroureterostomy)**, up to **90%** success; leak 10–24%
- Nephropexy → **left gains more**
- **Psoas hitch** 95–100%; spare **genitofemoral/femoral** nerves
- **Boari**: width **≥4 cm**, **≤3:1**, gaps to **12 cm**
- **TUU** (transureteroureterostomy) 90–97% (children 70%); ureterotomy **≥2 cm**
- TUU avoid: **urothelial cancer, stones, TB, radiation, RPF (retroperitoneal fibrosis), chronic pyelonephritis**
- Partial transection → **Heineke–Mikulicz**
- Robotic reimplant **~95%**; buccal **~87%**
- **Ileal ureter** 81–100%; acidosis ~20%
- **Yang–Monti** = less bowel → less metabolic
- **Autotransplant** = last resort; **5–10%** lost
- Follow-up: renogram/US **4–12 wk + 1 yr**

CLASSIC TRAPS

- Trap: **No haematuria excludes injury** — Actually: UPJ and pedicle injuries often have none
- Trap: **Hypotension reliable in children** — Actually: late, after ~50% volume loss
- Trap: **Explore every renal injury at laparotomy** — Actually: only expanding/pulsatile haematoma
- Trap: **Main vessel contained bleed = grade IV** — Actually: grade V in 2018
- Trap: **Extravasation needs a stent** — Actually: observe unless UPJ, obstruction, symptoms
- Trap: **Revascularise unilateral thrombosis** — Actually: observe if contralateral normal
- Trap: **Right renal vein ligatable like left** — Actually: right has no collaterals
- Trap: **Tunnelled reimplant in trauma** — Actually: non-tunnelled, refluxing
- Trap: **Bowel or autotransplant acutely** — Actually: delayed only
- Trap: **CT without delays excludes leak** — Actually: delayed phase mandatory